Website Traffic Forecasting - Analysis Report

Objective: analyse historical daily website traffic and forecast future visitor levels using ARIMA, SARIMA, Prophet and LSTM.

Data: 2167 daily records, 2014-09-14 to 2020-08-19 (expected 2167 days).

Validation: 0 missing values, 0 duplicate dates, 0 outliers flagged.

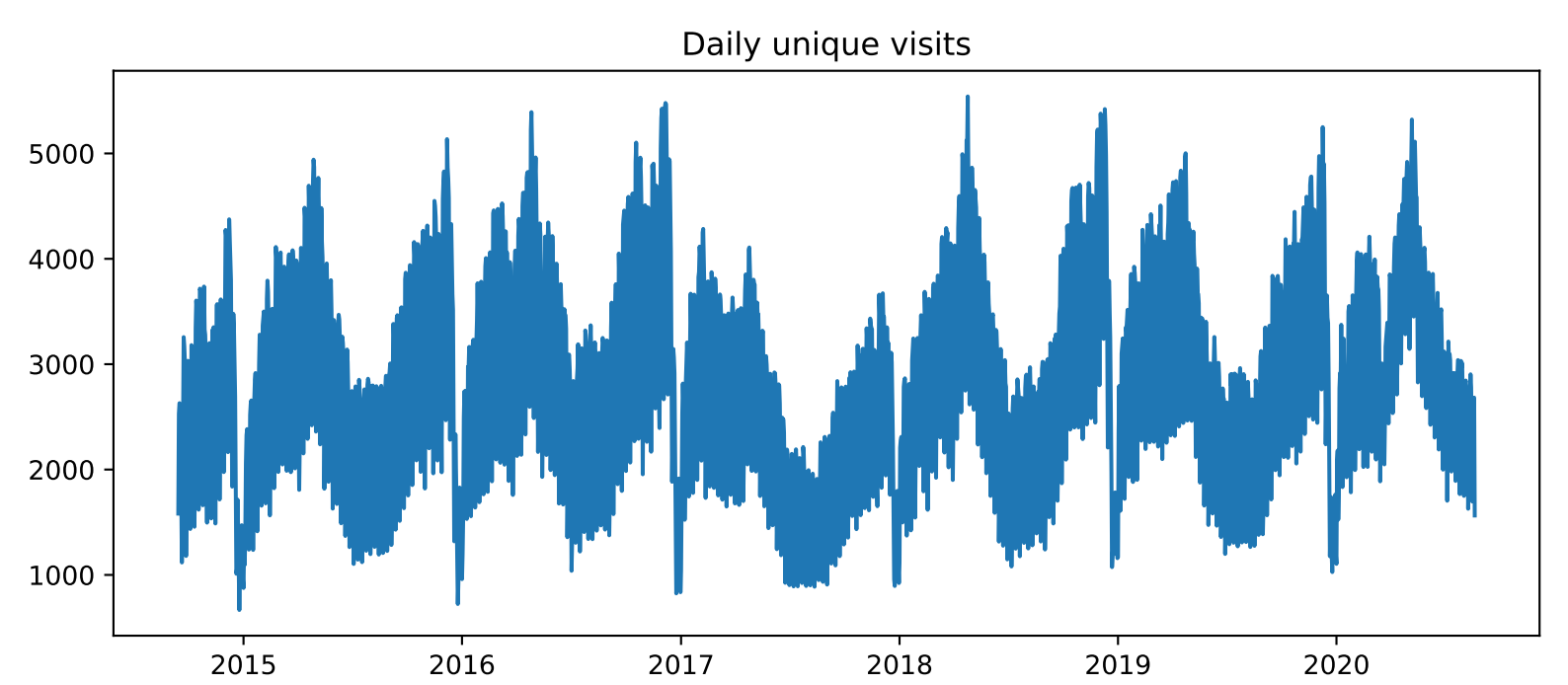
Train / test split: first 2077 days for training, last 90 days (2020-05-22 to 2020-08-19) for testing.

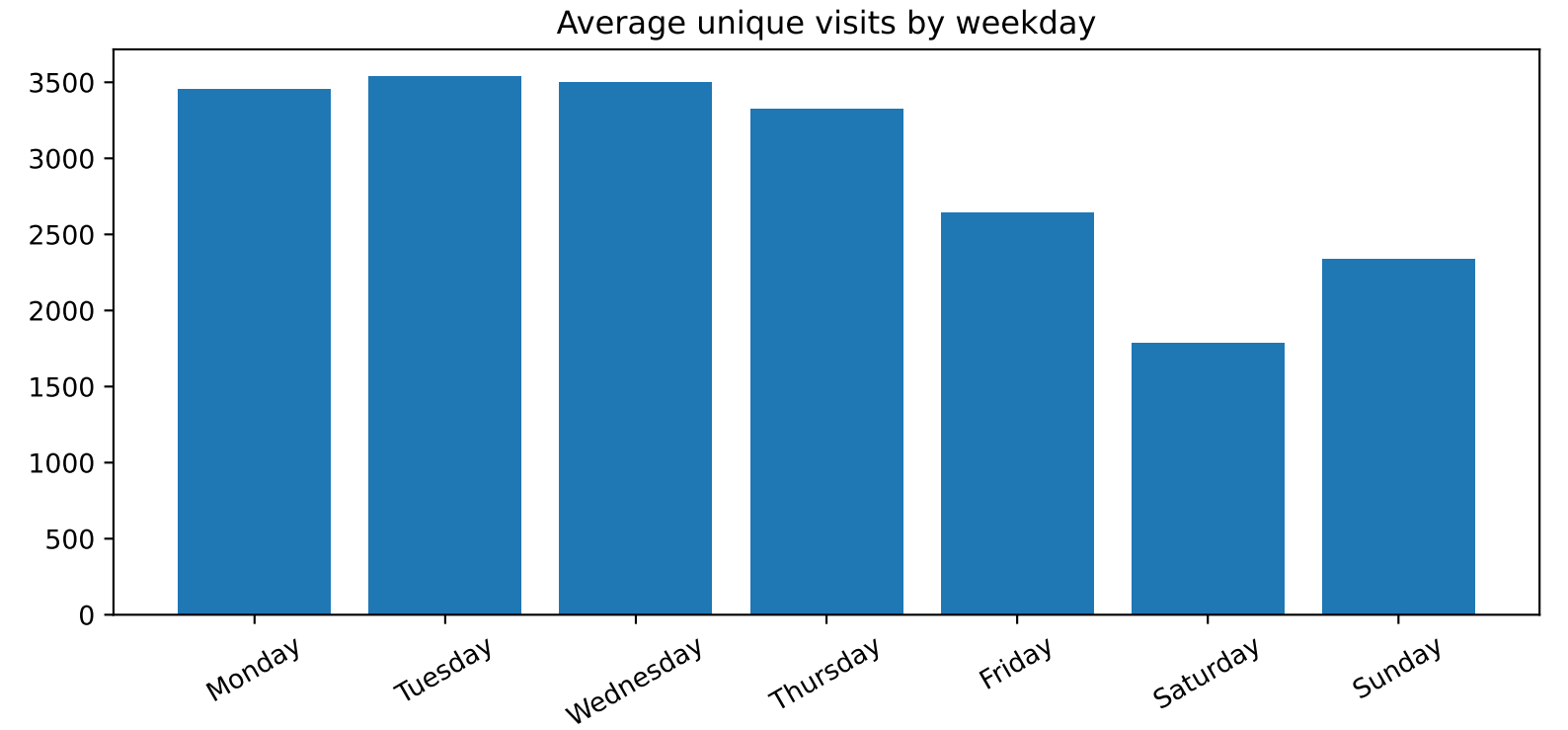
Best model by RMSE on the test days: SARIMA.

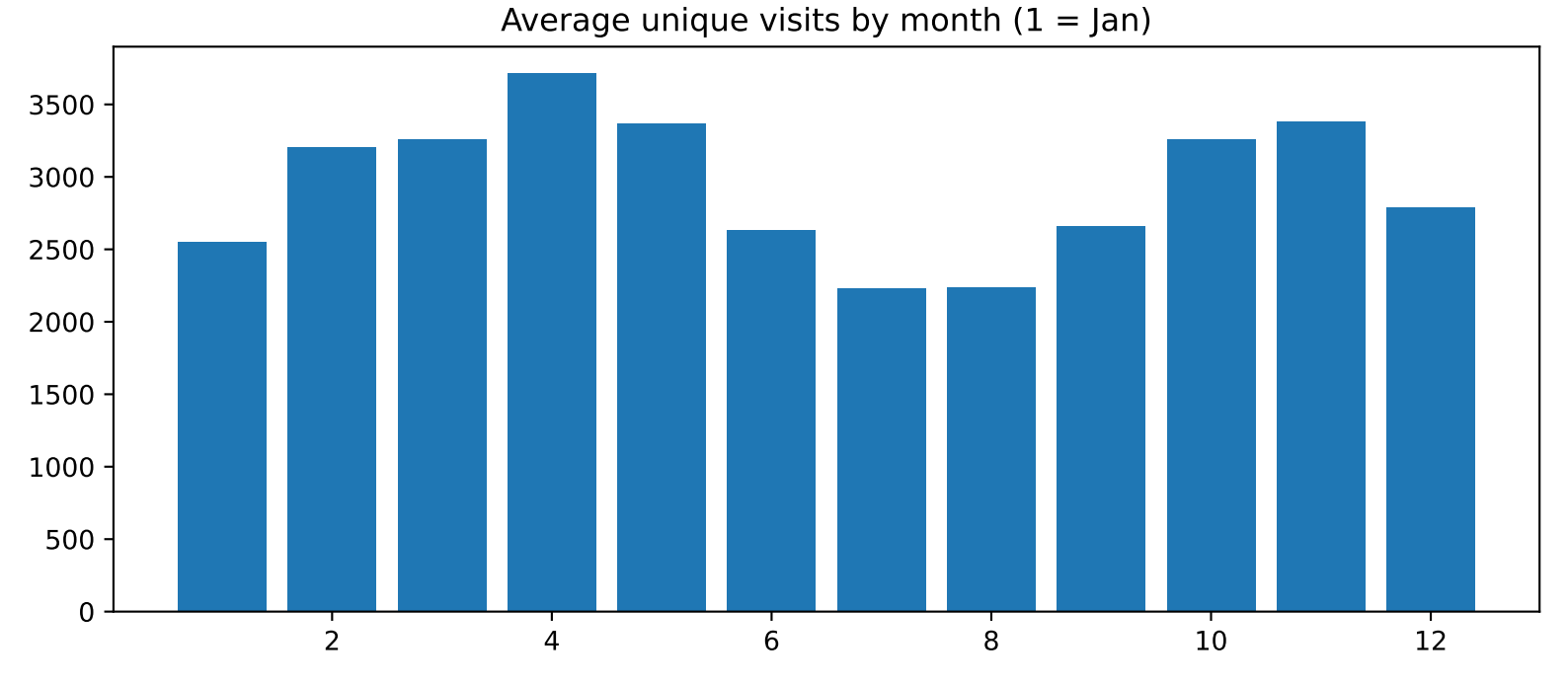
Metrics are measured in unique visitors per day (MAE, RMSE, MSE) and percent (MAPE).

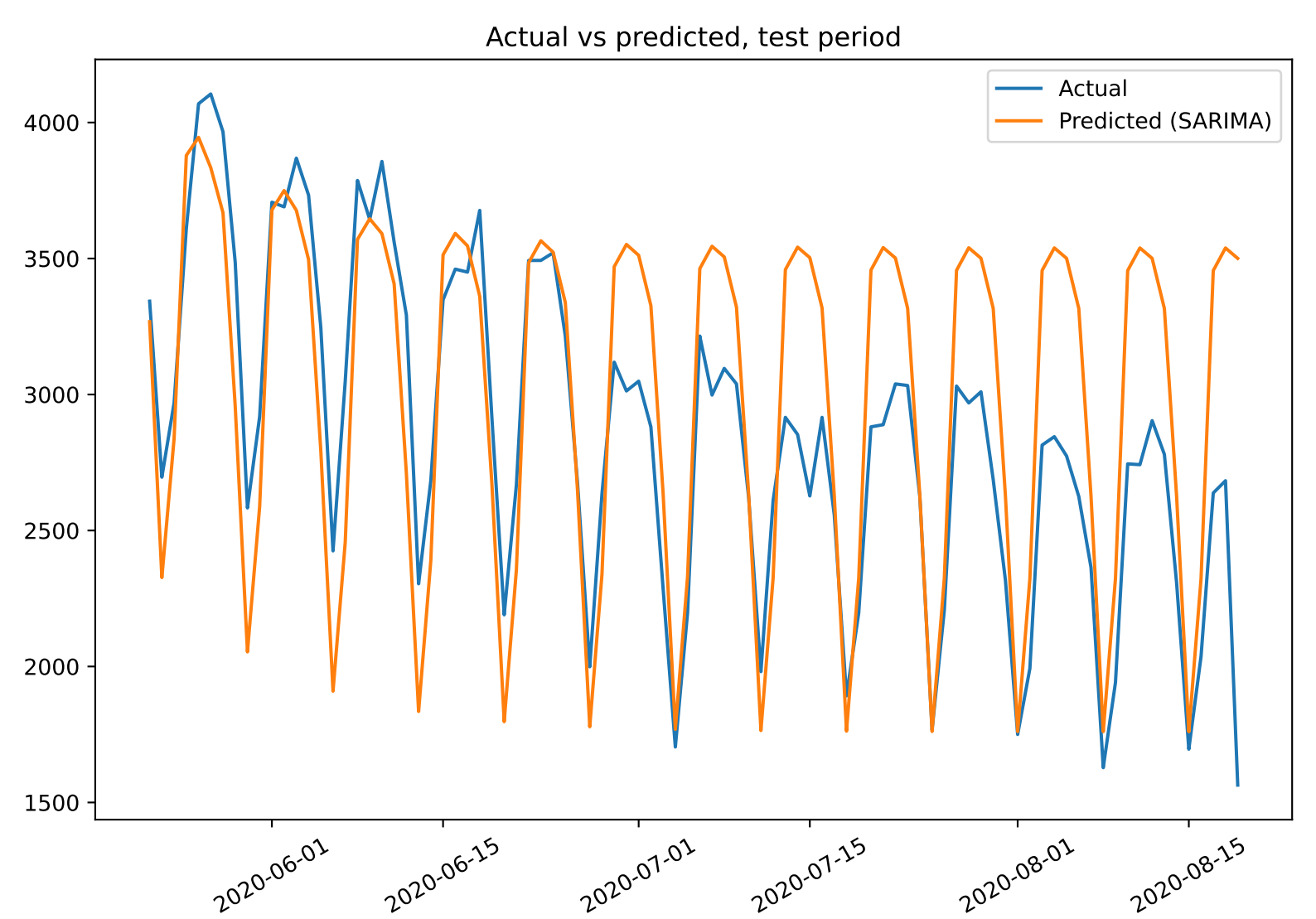
Model comparison (sorted by RMSE)

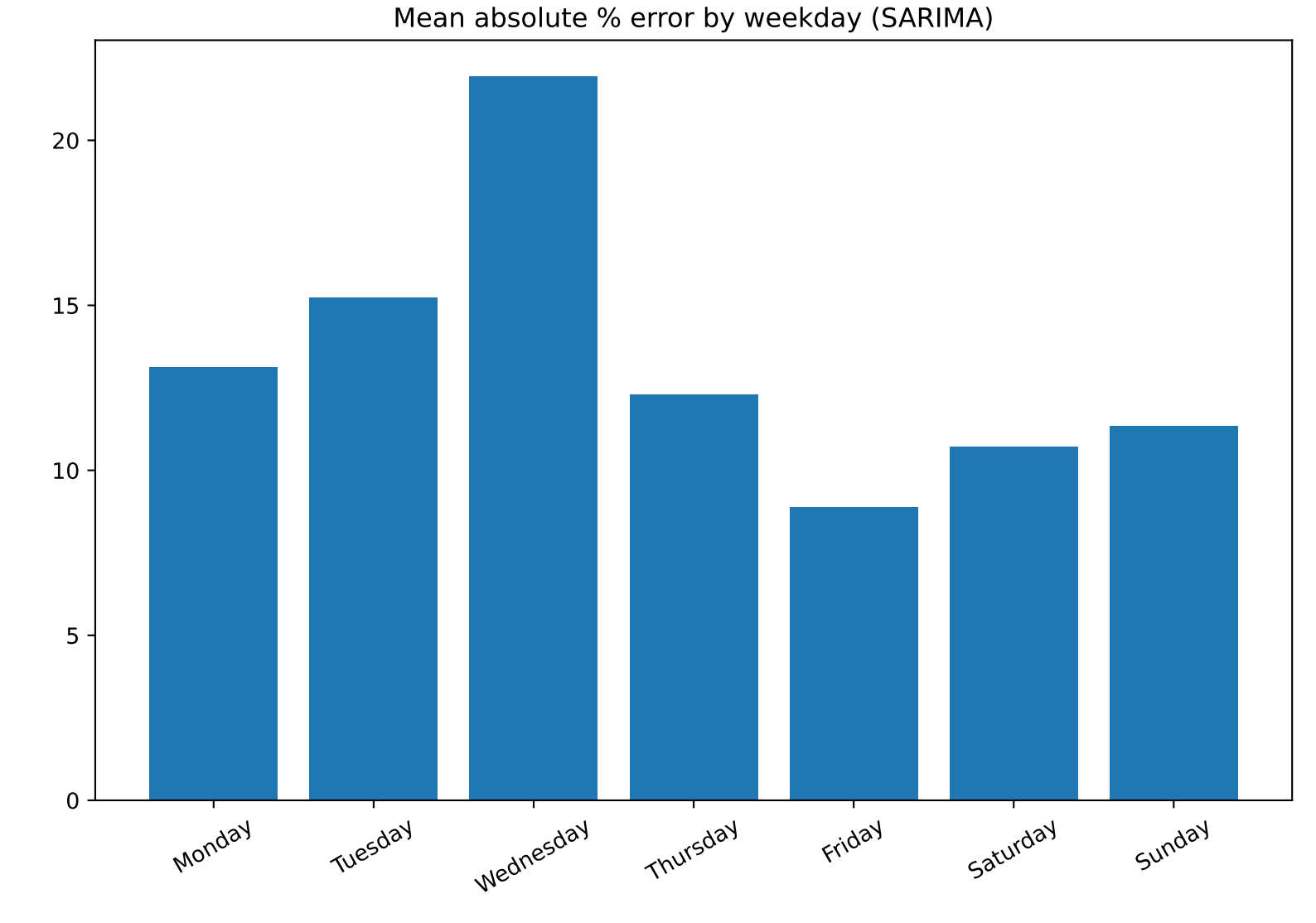
Model	MAE	MSE	RMSE	MAPE %
SARIMA	354.44	205761.3	453.61	13.37
Prophet	446.42	279571.84	528.75	17.34
Seasonal naive	945.44	1100619.13	1049.1	36.93
LSTM	940.72	1114307.01	1055.61	32.71
ARIMA	1218.88	1846973.96	1359.03	50.15

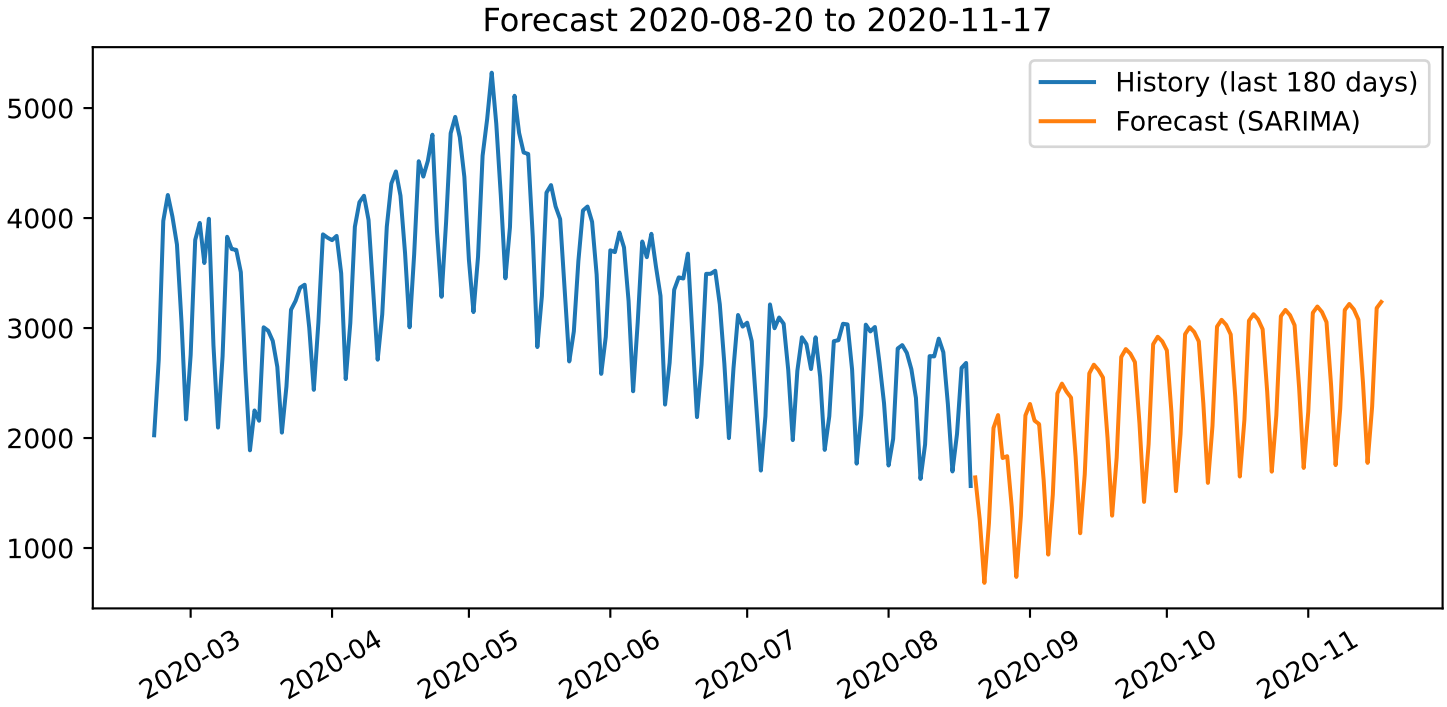

Daily unique visits
5000
4000
3000
2000
1000
2015
2016
2017
2018
2019
2020


Average unique visits by weekday
3500
3000
2500
2000
1500
1000
500
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Average unique visits by month (1 = Jan)
3500
3000
2500
2000
1500
1000
500
0
2
4
6
8
10
12


Actual vs predicted, test period
Actual
Predicted (SARIMA)
4000
3500
3000
2500
2000
1500
2020-06-01
2020-06-15
2020-07-01
2020-07-15
2020-08-01
2020-08-15


Mean absolute % error by weekday (SARIMA)
20
15
10
5
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday



Expected peak and low days in the forecast

Peak date	Peak visits	Low date	Low visits
2020-11-17	3237.0	2020-08-22	683.0
2020-11-10	3219.0	2020-08-29	737.0
2020-11-03	3196.0	2020-09-05	940.0
2020-11-16	3182.0	2020-09-12	1134.0
2020-11-11	3169.0	2020-08-23	1226.0

Limitations:

- Data is daily only; hourly forecasting is not possible with this dataset.
- Sessions are not recorded; unique visits is used as the closest proxy.
- The data covers a single website.
- Models were compared on one 90-day test window.
- Forecast dates follow the end of the dataset, so they cannot be checked against real data.